A Galaxy Law Tied to Dark Energy, and a Map of What It Leaves Unexplained

Flagship working note (PAPER67), DRAFT, 2026-11-01, for the owner's review; not deposited. AI-assisted research programme; not peer reviewed. $\kappa = \frac{1}{2}$ is fitted. The theory is not closed. Nothing here says the data favour this framework over any other.

Abstract

We report what is left of a research programme after its own final grade. A galaxy law, $g_N = g^2/(g + a_0)$ with $a_0 = \kappa c\sqrt{G\rho_{\rm DE}}$ and $\kappa = \frac{1}{2}$ fitted, describes isolated disc galaxies well on the canonical footing: SPARC rms 0.0541 dex at median Υ 0.52, no tilt with speed, no slope with surface brightness at fixed speed. The law is a restatement of MOND's simple form with a proposed origin for its scale. Earlier authors proposed that origin, and the scale's tie to dark energy, as distinct from cH_0, has no decisive test yet.

The law fails the Milky Way's vertical force, IC 1613, several Local Group classes, cluster cores and the unscreened Solar System, and at 60–150 km s^{-1} it asks for about twice the stellar mass estimated from photometry. It has no action and supplies no dust-like term for the CMB. To cover those we developed a cold dark energy sector. On measured data all four candidates built on it fail, and the sector is now retired. We publish the scorecard, the two walls that stopped it, and twelve requirements any successor must meet. This is numerical evidence for the families tried, not a no-go theorem. We close with dated tests that can still kill the law.

1 Scope

This is a programme report, not a complete theory. Three owner decisions on the record set its form: the a_0 floor was removed from the cold sector (2026-10-31), the search for a derived threshold stopped (2026-11-01), and the cold sector was retired (2026-11-01). Earlier deposits stand as dated records. The two footings, canonical $a_0 = 9.3603 \times 10^{-11}$ and alternative 1.1312×10^{-10} m s^{-2}, are never pooled. This paper's audit script checks the quoted numbers against committed files. It doesn't check the prose.

2 The law, and whose idea it is

$$g_N = \frac{g^2}{g + a_0}, \qquad a_0(z) = \kappa c\sqrt{G\rho_{\rm DE}(z)}, \qquad \kappa = \tfrac{1}{2}\ \text{(fitted)}.$$

Credit. The first relation is Milgrom's modified dynamics in its simple form [1]. The numerical nearness of a_0 to cH_0 and to $c^2\sqrt{\Lambda}$ is his observation [1, 2]. Limbach, Psaltis & Özel [3] proposed the tie to the dark energy density. None of this originated here.

κ isn't derived. Of 537 attempts on the record, none derives $\kappa = \frac{1}{2}$. It isn't singled out by SPARC either. A profile likelihood over 175 galaxies has a free best fit of 1.0766×10^{-10}. Against it, $\kappa = \frac{1}{2}$ on $\rho_{\rm DE}$ reads $\Delta\chi^2$ 63.90 (1.82σ with galaxy-clustered errors), the alternative footing 7.04 (0.60σ), and Milgrom's own $cH_0/2\pi$ 5.34 (0.53σ). So Milgrom's value fits at least as well as ours.

What this programme adds: a reconstruction of $a_0(z)$ from raw BAO and supernovae with a stated ruler, the graded comparison below, and the failure record.

The band, and what it rests on. The reconstruction gives $a_0(z)/a_0(0)$ of 0.91 [0.76, 1.03] at $z = 1$, 0.94 [0.71, 1.12] at $z = 1.5$ and 0.91 [0.33, 1.21] at $z = 2.33$ (68 %), against 1.76, 2.35 and 3.49 for the rival $a_0 \propto H(z)$. Constant dark energy isn't rejected (1.97σ, primary). **This band isn't free of the retired sector's input:** it takes $\rho_m(z) = \omega_m(1 + z)^3$ with a cold mass from the CMB, and with positivity alone, no CMB and no H_0, its check fails. The law doesn't explain that matter term (Section 5).

The distinctive claim is untested. At $z \approx 0$ the law can't be told from $a_0 \propto cH_0$. Separating them at 3σ needs a_0 to 0.055 dex at $z = 1$ and 0.084 dex at $z = 2.33$. The best public sample reaches 0.4–0.5 dex.

Table 1: The law on isolated systems. FG1 is the frozen final grade; a clean-room regrade (CRG1) reproduced its table, 289 match, 6 close, 0 disagree, mainly a test of code.

line	canonical	alternative	lane
SPARC rms, median Υ (line 0.50–0.80)	0.0541, 0.52	0.0537, 0.44 **fails the line**	FG1
tilt with speed	+0.02	−0.02	FG1
LITTLE THINGS, median E	−0.025	−0.064	FG1
63 groups, Z	+0.31	−0.31	FG1
slope with surface brightness, 60–120 km s^{-1}	-0.016 ± 0.061, flat	-0.023 ± 0.063, flat	HS1
local a_0, MeerKAT against SPARC	+0.032 dex (line 0.15)		PAPER60

3 What it does in isolated galaxies

A caution on the Υ line. The 0.50–0.80 line was first written for this law, and its lower edge is a population model's central value, not a measurement. So passing it is weaker than it looks; Section 4 sets it against measured masses.

Surface brightness is the strongest result. At fixed speed, a halo set by mass alone can't follow the disc: every version of the retired sector's halo tilts, −0.17 to −0.29 per dex, and the law doesn't. That's a real discriminator on the data used, between the law and these halos only.

Isolated-galaxy lensing is shape, not a clean amplitude. KiDS-1000 prefers the law's shape over the retired halos ($\Delta\chi^2$ +366.0 against the best native halo). But the law's amplitude depends on colour and on the baryon inventory: blue lenses read $A = 1.016 \pm 0.086$ in one lane and 0.878 in a later forward model that adds a hot gas inventory, and red lenses 1.587 and 1.313. No tested model fits these profiles in absolute terms.

4 Where the law fails

- **Milky Way.** With published baryon priors, 7 of 175 cells pass the frozen line (alt 4), all with a short disc and the lightest local column. Those cells have $K_z(R_0)$ 76.7–80.9 against 66.4 ± 2.2, +4.7 to +6.6σ, and leave the outer slope excluded (Z −3.88). FG1 counts this as FAIL.
- **IC 1613.** $E = -0.593$, $Z = -7.9$; it fails under every halo-free inclination. One galaxy: disfavoured, not excluded.
- **Local Group classes.** Z: MW ultra-faints +5.22, bright M31 +2.20, other M31 +2.29, MW classicals +1.69, field +0.57. The path has no external-field term. An external field lowers the boost and these are under-predictions, so omitting it doesn't produce them; that wasn't computed.
- **Measured stellar masses, 60–150 km s^{-1}.** The law's Υ is 0.62 and 0.65 where GSWLC stellar masses give 0.277 and 0.304 (Z +3.4, +5.6). These are estimators against W1 luminosity, not a 3.6 μm measurement. Gas dominance doesn't excuse the 100–150 bin.
- **Disc weighings.** DiskMass σ_z reads $r = 1.250$ on the one-zone estimator and 1.342 with the exact integral, outside 0.90–1.10; those lanes are frozen inconclusive, so FG1 doesn't grade it. Same-population dispersions [5, 6, 7] correct r by c of 1.414–1.517, giving 0.824–0.884 one-zone and, with exact Jeans and measured thickness, inside the line. But the sources' own tables run c from about 1.24 at the relevant radius to 1.88 in integrated light, so the question is open (OD1, NOT SEPARATED). NGC 628 reads $\Upsilon_{3.6} = 0.75 \pm 0.18$ and NGC 6946 about 0.4.
- **The Solar System.** The unscreened law fails Cassini's Q_2 by 2.44–4.59 times the 2σ line (alt 2.85–5.19). The pass in earlier reports belongs to a screened covariant theory. That line's final verdict (v3s) was that it holds as a galaxy law and fails as gravity plus cosmology. So the law as written here needs a screen it doesn't contain.

5 What it leaves unexplained

- **Cluster cores.** From measured gas, in 318 eROSITA clusters with raw KiDS shear, the law is short by $A = 1.91 \pm 0.12$ at 0.5–1.2 Mpc (alt 1.75 ± 0.11) and matches at 1.2–3 Mpc, $A = 1.01 \pm 0.09$. It fails 6 of 15 keys on an older template (alt 10) and reads -0.151, rms 0.205 on X-COP (numbers only).
- **Outer shear against gas mass.** $\Delta\Sigma_{\rm OUTER} \propto M_{\rm gas}^{s}$ with $s = 0.90 \pm 0.21$, against the law's 0.47 and Newton's 0.84. The amplitude agreement above is an average over a slope the law misses.
- **The CMB and growth.** a_0 is absent from the linear FRW equations, so the law can't act there. The record has no CMB pass without a dust-like component. With one, a window exists at $z_{\rm on} \in [3 \times 10^4, 10^6]$ on the channels run. Retiring the sector leaves this unanswered. **The law offers no replacement, and we don't claim it needs none.**
- **The matter term.** The law takes $\rho_{\rm DE}$ from an expansion history whose matter term is several times the baryons. That term is unexplained here.
- **An action.** None is selected. The covariant attempt (v3s) is closed, and the a_0 sector's action stays open.

6 The cold sector: tried and retired

What it was. A gravitating component with, at linear order, $w = 0$, $c_s = 0$, geodesic motion and an abundance set by an integration constant. A referee will read those as the linear behaviour of cold dark matter, and that's correct; it was never proposed as a particle, and the programme's no-particle rule is unchanged. Its candidates: C1, the native collapse rule N; C2, N with the measured gravitational response; C3, a basin split at $T \approx 10^{12.3}\,M_\odot$; with C4, the law alone, as the control.

Table 2: FG1, canonical. P pass, F fail, – not graded, which isn't a pass. FG1 declares no winner, because the lines aren't independent.

	S	Tl	LT	G	K	X	Ln	Dw	MW	DM	IC	alt
C1 native N	P	F	F	P	F	P	F	F	F	F	F	no a_0, same
C2 N + response	F	F	F	P	F	F	F	–	F	P	–	no a_0, same
C3 basin split	P	P	P	P	F	P	P	P	F	–	F	also S, Dw fail
C4 law alone	P	P	P	P	F	–	P	F	F	–	F	also S fails

S SPARC, Tl tilt, LT LITTLE THINGS, G groups, K 15 cluster keys, X X-COP, Ln KiDS isolated, Dw Local Group, MW Milky Way, DM DiskMass, IC IC 1613. C4's K is on an older template than C1's and C3's. FG1's tag [GAS-CONDITIONAL]: at X-COP's gas scale +0.111 the native rule reaches 15 of 15 keys, rescuing K only; the measured scale is $+0.005 \pm 0.017$.

Wall 1: spirals at 150–250 km s^{-1}. Whatever is dust in the CMB falls into galaxies, and a live disc measures it contracting: $f_Q = 0.872$ [0.54, 1.07] (LC1), 0.946 [0.78, 1.08] with the disc grown from turnaround (PZ1). The SPARC side needs net f of about 0–0.2 at the 0.50 Υ line (RM2, six of 75 cells, three fitted numbers). This is the known Tully–Fisher zero-point problem [4] in measured numbers. A cored halo's response is somewhat weaker, $f_B = 0.753$ [0.729, 0.777] (DR8), not enough. Whether the 0.50 line is itself too high is open (Section 4, UL1, OD1); moving it wouldn't clear tilt or LITTLE THINGS.

Wall 2: outer cluster shear on measured gas. G-OUT-M3 reads 1.56 against 1.04, $Z = +4.41$ statistical, about +3.03 to +3.51 with systematics (M3A1). It's not an artefact on anything measurable from disk, nor the model's edge, and native two-halo supplies 3–11 % of +21 to +27 %.

No derived switch. The basin split's T was located, never derived. Three frozen attempts failed: 13.94 for the turned-around mass and 13.54–13.89 for a clock, against a window of 12.19–12.39 in $\log M$, and a candidate action had no link to mass. In shift-symmetric $P(X)$, a turn-on has $c_s^2 < 0$ at all 14 grid points, a result within that family (BOLTZ1).

Status. Retired by the owner on 2026-11-01. Two lanes frozen before that decision are still running and will be recorded here. What survives from this work as fact, not as a theory: native collapse passes X-COP on hydrostatic masses, +0.013, rms 0.075, and one adiabatic response has the opposite sign in discs and cluster cores.

Table 3: Doors tried in the closing push, by each lane's own label. Full list: the closing ledger.

idea	lane	result
density term in the law	DR1	SPARC BREAKS, CLUSTERS NOT REACHED
phase-space cap on response	DR7	BITES, FAILS [NOISE-LIMITED]
retention replaces the mass line	RET1	DERIVED LINE FAILS, FITTED ϵ FAILS
barotropic halo	HB1	MOVES, FAILS
fresh hunt, 19 routes	MH1	no mechanism survives whole
starting point	SP3	no response-on start reaches Υ's line in the owner's grid
symbolic search	SL1	TRAIN ONLY, 0 survivors

7 What a successor must do

Table 4: Requirements from combining the failures. Older rows are on older chains, and FG1 supersedes any it covers. Coverage: an inventory that opened about 90 of about 330 lanes.

#	must	law	native halos
1	keep galaxies tight	**0.0541**	0.0682 / 0.0694
2	stay flat with surface brightness	-0.016	-0.173
3	sit on Υ at every speed	0.52, alt 0.44	0.75, running 1.20 to 0.57
4	show little net contraction in spirals	not needed	needs $f \leq 0.2$, measured 0.872
5	fit dwarfs	-0.025	$+0.138$; 0 of 5 classes
6	fit IC 1613	-0.593	-0.474
7	fit the Milky Way	K_z $+4.7$ to $+6.6\sigma$	asks 4e10 $M_\odot$ inside R_0
8	fit isolated lenses	shape preferred	$+366.0$
9	add mass in cluster cores	A 1.91	G-IN -0.49, G-MID $+2.17$
10	match outer cluster shear	slope 0.47 against 0.90	G-OUT-M3 $+4.41$
11	act as dust in the CMB	no cold term	$z_{\rm on} \in [3 \times 10^4, 10^6]$
12	have an action	open	$P(X)$ dust limit only

The collisions. Rows 4 and 11: dust in the CMB falls into galaxies. Rows 1–3 and 10: the amount needed runs from about 0.8 of native on SPARC to 1.22–1.80 on clusters. Rows 7 and 8: systems of the same mass want different things. Selectors closed on inputs tested: potential, σ, acceleration, environment, formation epoch, gas state, bars. None of this is a theorem.

One family left outside. A 5–11 eV fermion would address rows 4 and 11 directly. Its forest kill isn't on the current footing: a 1-D proxy against a ΛCDM yardstick falls 1000–5000 times short, undecided on raw forest data. Other liabilities stand, and it wouldn't fix rows 6 or 7. It's outside the no-particle rule.

8 Dated tests, and their real power

- **ALMA U4_28473, public 2026-11-06 02:42:31 UTC** (PAPER66 v1.2, DOI 10.5281/zenodo.23868979). It tests this galaxy's mass budget, not $a_0(z)$, and one disc excludes no model.
- **GS4_24110,** [CI](2–1), 2026-11-28 by earlier metadata, to be rechecked. Every $n = 1$ against $n = 2$ pair is at or below 0.51σ, so it can't separate them.
- **Gaia DR4, about December 2026** (pre-registered 2026-07-16, with later amendments). Wide binaries: Newton 1.000, an ungated MI band 1.05–1.10, AQUAL point 1.137, MOND's benchmark 1.33; the DR3 dry run's 1.205 ± 0.035 is evidence for nothing. An amendment registers a gated branch for which 1.000 inside 30 kAU is also predicted, so that branch needs its own test beyond 50 kAU, about 2000 clean pairs per bin. The solar-acceleration dipole's kill bands are void under the amended theory, so that front returns no verdict. *Reading, not a frozen line:* these bands were written for the action-carrying framework. The bare law here contains no gate, so a Newtonian result at 2–30 kAU would count against it.
- **SP1, SP2 sealed lensing predictions.** Scored as frozen, including the retired route's rows.

- **Named, not scheduled:** one clean deep-MOND a_0 point at $z \approx 2.5$ to about 0.10 dex. That's the measurement that would actually test the dark energy tie.

9 Method

Each lane commits frozen criteria alone before any script or number, runs a MUTATE twin that must fail, reruns byte-identically and publishes failures under frozen labels. Most lanes were not blind. Grading was done by the same programme that proposed the models; the clean-room regrade tested code, not independence of definitions. No outside group has reproduced these results.

Not claimed

A complete, closed or unified theory. That the data favour this framework. κ, T or any abundance as derived. That the tie of a_0 to dark energy originated here. That the requirements list is a no-go theorem.

Data and code

All lanes named here are in `rebuild_2026_10_22/` and `campaign_fresh_gravity/`. Scorecard: `FG1_final_grade`. Ledger: `CLOSING_LEDGER_2026-11-01_DRAFT.md`. Audit: `PAPER67_audit.py`.

References

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[5] S. Aniyan et al., MNRAS 456, 1484 (2016), arXiv:1511.06047.

[6] S. Aniyan et al., MNRAS 476, 1909 (2018), arXiv:1802.00465.

[7] S. Aniyan et al., arXiv:2010.03991 (2020).